

[447] ON THE RATIONAL TRANSFORMATION BETWEEN TWO SPACES.

(continued from book page 203 / scan page 225.)

[Book pages 204–211 are occupied almost entirely by Cremona’s tables of principal systems for the rational transformations of orders $n = 2, 3, 4, 5, 6, 7, 8, 9, 10$ and of the generic types $n = p$, $n = 2p$, $n = 2p + 1$, $n = 3p$, $n = 3p + 1$, $n = 3p + 2$, $n = 4p$, $n = 4p + 1$, $n = 4p + 2$, $n = 4p + 3$. Each table is a rectangular grid: the top row lists the columns of partitions, indexed by an outer label n ; the body shows the integers $a_1, a_2, \dots, a_{n-1}$ that constitute the partition. The tables are sibi-reciprocal, the conjugate pairs being marked a_r and a'_r . In what follows I retype the body text and reproduce the smaller auxiliary tables verbatim; for the long Cremona tables of article 41 on book pp. 204–207 I give a faithful but compact representation, preserving every entry as it stands in the printed page.]

Article 41 (continued): Cremona’s Tables of Partitions.

The tables are read as follows: in each block the topmost row gives the column-label n (the order of the transformation), then under it stand successive columns, each headed by the values $a_1, a_2, \dots, a_{n-1}$ written one below the other. A blank entry denotes 0.

Table $n = 7, n = 8, n = 9, n = 10$. (Book p. 204.)

Block I, $n = 7, 8, 9$:

	$n = 7$						$n = 8$								$n = 9$									
	3	4	6	8	10	12	14	3	1	4	6	8	12	12	16	4	2	0	3	7	1	3	0	1
a_1	3	4	6	8	10	12	14	3	1	4	6	8	12	12	16	4	2	0	3	7	1	3	0	1
a_2		1	3	3	5	5		1	3	6	0	2	3	1		2	4	0	3	7	1	3	0	1
a_3			1	1	2	3	2		1	0	5	0	0	0		0	1	0	0	0	0	0	0	0
a_4				1		1	1	1			0	5	1	1				0	0	0	0	0	0	0
a_5					1	1	1						1					0	0	0	0	0	0	0
a_6							1											0	0	0	0	0	0	0

Block II, $n = 10$. Last column marked by Cremona as “omitted by Cremona.”

	18	5	1	0	0	3	6	2	4	1	2	1	3	3	3	0	1
a_1	18	5	1	0	0	3	6	2	4	1	2	1	3	3	3	0	1
a_2		4	2	0	3	0	3	1	3	3	0	6	1	0	0	0	0
a_3			2	3	0	1	0	2	0	5	0	1	0	5	0	0	0
a_4				2	3	0	1	0	2	1	3	0	0	0	5	0	1
a_5						1	0	2	1	0	2	0	0	0	0	5	0
a_6							1	1	0	0	1	0	0	0	0	0	0
a_7								1	0	0	0	0	0	0	0	0	0
a_8									0	0	0	0	0	0	0	0	0
a_9		1								0	0	0	0	0	0	0	0

[Footnote on p. 204: “Omitted by Cremona.”]

Tables for the generic forms $n = p, n = 2p, n = 2p + 1, n = 3p$. (Book p. 205.)

Form $n = p$.

	a_1	a_{p-1}
$a_1 = 2p - 2$	$2p - 2$	1
$a_{p-1} = 1$	1	1

Form $n = 2p$.

	a_1	a_2	a_{p-1}	a_p
$a_1 = 3$	3			$2p - 2$
$a_2 = 2p - 2$		$2p - 2$	1	
$a_{p-1} = 1$		1		
$a_p = 4$	$2p - 2$			1

Form $n = 2p + 1$.

	a_1	a_p	a_{p+1}	a_{2p-1}
$a_1 = 3$	3			$2p - 1$
$a_2 = 2p - 1$				
$a_p = 4$	$2p - 1$	1	1	
$a_{p+1} = 1$	4	1		
$a_{2p-1} = 1$		$2p - 3$	1	1

Form $n = 3p$.

	a_1	a_2	a_p	a_{p+1}	a_{2p-1}
$a_1 = 4$	4				$2p - 2$
$a_2 = 2p - 2$		$2p - 2$			
$a_p = 3$					
$a_{p+1} = 4$	$2p - 2$				4
$a_{2p-1} = 1$					1
$a_{3p-2} = 1$	1	$2p - 2$	1	4	1

Tables for the generic forms $n = 3p + 1$, $n = 3p + 2$, $n = 4p$, $n = 4p + 1$, $n = 4p + 2$, $n = 4p + 3$.
(Book p. 206.)

Form $n = 3p + 1$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2$	2	$2p - 2$		
$a_2 = 7$	7			$2p - 1$
$a_p = 1$		1		
$a_{2p-1} = 3$	$2p - 1$		1	3

Form $n = 3p + 2$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p - 4$	$2p - 4$	1		$2p - 1$
$a_2 = 7$		3	4	1
$a_p = 1$			1	3
$a_{p+1} = 3$	$2p - 2$		1	4
$a_{2p} = 1$				1

Form $n = 4p$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 9$	9	6	2	$2p - 1$
$a_2 = 3$	3		3	5
$a_3 = 3$		$2p - 3$	1	2
$a_4 = 2p - 2$	1		$2p - 1$	1
$a_p =$				
$a_{p+1} =$				
$a_{2p+1} = 1$				1
$a_{4p-1} = 1$	4	1	1	

Form $n = 4p + 1$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2$	2	$2p - 2$	3	$2p - 1$
$a_2 = 3$	3		4	7
$a_3 = 2p - 3$		1		4
$a_4 = 2p - 3$				1
$a_p =$	1	3		
$a_{p+1} =$				
$a_{4p-3} = 1$			1	

Form $n = 4p + 2$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p - 4$	1	$2p - 2$	1	$2p - 1$
$a_2 = 7$	3		3	4
$a_3 =$	2		4	3
$a_4 = 2p - 4$		$2p - 2$		4
$a_p =$			3	1
$a_{p+1} =$				
$a_{4p-3} = 1$			1	1

Form $n = 4p + 3$.

	a_1	a_2	a_p	a_{2p-1}
$a_1 = 2p - 2$	9	$2p - 3$	2	$2p - 1$
$a_2 = 3$	6		3	5
$a_3 = 3$		$2p - 3$	1	2
$a_4 = 2p - 2$	1		$2p - 1$	5
$a_p =$				1
$a_{p+1} = 3$		$2p + 1$	1	1
$a_{2p+1} =$	1		$2p + 3$	1
$a_{4p-1} = 1$		$2p + 1$	1	1

42. The system $(a_1, a_2, \dots, a_{n-1})$ geometrically determines completely the system $(a'_1, a'_2, \dots, a'_{n-1})$; it ought therefore to determine it arithmetically; that is, given the one series of numbers, we ought to be able to determine, or at least to select from the table, the other series of numbers. Cremona has shown that the two series consist of the same numbers in the same or a different order. By examination of the tables, it appears that there are certain columns which are single (that is, no other column contains in a different order the same numbers), others that occur in pairs, the two columns of a pair containing the same numbers in a different order. Where the column is single, it is clear that this must give as well the values of $(a'_1, a'_2, \dots, a'_{n-1})$ as of $(a_1, a_2, \dots, a_{n-1})$. Where there is a pair of columns, as far as Cremona has examined, if the one column is taken to be $(a_1, a_2, \dots, a_{n-1})$ the other column is $(a'_1, a'_2, \dots, a'_{n-1})$; it appears, however, not to be shown that this is universally the case; viz., it is not shown but that the two columns, instead of being reckoned as a pair, might be reckoned as two separate columns, each by itself representing the values as well of $(a_1, a_2, \dots, a_{n-1})$ as of $(a'_1, a'_2, \dots, a'_{n-1})$; neither is it shown that there are not, in any case, more than two columns having the same numbers in different orders. It seems, however, natural to suppose that the law, as exhibited in the tables, holds good generally; viz., that the tables contain only single columns, each giving the values as well of $(a_1, a_2, \dots, a_{n-1})$ as of $(a'_1, a'_2, \dots, a'_{n-1})$; or else pairs of columns, one giving the values of $(a_1, a_2, \dots, a_{n-1})$, and the other those of $(a'_1, a'_2, \dots, a'_{n-1})$; or, say, that the partitions are either *sibi-reciprocal*, or else *conjugate*.

43. Assuming that the two systems $(a_1, a_2, \dots, a_{n-1})$ and $(a'_1, a'_2, \dots, a'_{n-1})$ are each known, there is still a question of grouping to be settled; viz., the Jacobian of the first plane consists of a'_1 lines, a'_2 conics, ... a'_{n-1} unicursal $(n - 1)$ -thics; each line, each conic, &c., passes a certain number of times through certain of the points $a_1, a_2, \dots, a_{n-1}$: but through which of them? For instance, each of the a'_1 lines will pass through two of the points $a_1, a_2, \dots, a_{n-1}$: will these be points a_1 , or points a_2 , &c., or a point a_1 and a point a_2 , &c.? The mere symmetry of the different groups of points determines certain conditions of the solution¹; for instance, if any particular one of the a'_1 lines passes through two points a_1 , then each of the a'_1 lines must pass through two points a_1 ; and since the points a_1 are symmetrical, we must in this way use all the pairs of points a_1 ; that is, if $a'_1 = \frac{1}{2}a_1(a_1 - 1)$, but not otherwise, it may be that each of the a'_1 lines passes through two of the points a_1 . In the case of an equality $a_1 = a_2$ we could not hereby decide whether the line passed through two points a_1 or through two points a_2 . So, again, if any one of the a'_1 lines pass through a point a_1 and a point a_2 , then each of the a'_1 lines must do so likewise, and we must hereby exhaust the combinations of a point a_1 with a point a_2 ; viz., the assumed relation can only hold

¹It is by such considerations of symmetry that Cremona has demonstrated the before-mentioned theorem of the identity of the numbers $(a_1, a_2, \dots, a_{n-1})$ and $(a'_1, a'_2, \dots, a'_{n-1})$.

good if $a'_1 = a_1a_2$. Similarly, each of the a'_2 conics will pass through five of the points $a_1, a_2, \dots, a_{n-1}$; each of the a'_3 nodal cubics will pass twice through one (have a double point there) and through six others of the points $a_1, a_2, \dots, a_{n-1}$;—which are the points so passed through? I do not know how a general solution is to be obtained, but most of the cases within the limits of the foregoing table have been investigated by Cremona.

The results may conveniently be stated in a tabular form; the tables exhibit in the outside upper line the values of $a_1, a_2, \dots, a_{n-1}$, and in the outside left-hand line the values of $a'_1, a'_2, \dots, a'_{n-1}$, and they are to be read as

a'_1 lines

a'_2 conics

&c.

}

passes () times through () of the points $a_1, a_2, \dots, a_{n-1}$

respectively; the numbers in the table being those of the points passed through, and the indices in the table (index = 1 when no index is expressed) showing the number of times of passage, that is, showing whether the point is a simple, double, triple, &c., point on the curve referred to.

44. Thus (in the tables which follow) the last of the tables $n = 6$ gives the constitution of the Jacobian of the first plane, where the principal system is $(3, 4, 0, 1, 0)$; and it is to be read:

Each of the 4 lines passes through 1 of the points a_1 and through the point a_4 ;
The 1 conic „ „ 4 of the points a_1 and through the point a_4 ;
Each of the 3 cubics „ „ 2 of the points a_1 , 4 of the points a_2 , and twice through the point a_4 (that is, a_4 is a double point on each cubic).

It is hardly necessary to remark that the tables are sibi-reciprocal, or else conjugate, as appears by the outer lines of each table.

Table $n = 2$.

	a_1 $\frac{1}{3}$
$a'_1 = 3$	2 [was originally printed, 3.]

Table $n = 3$.

	a_1 $\frac{4}{4}$	a_2 $\frac{1}{1}$
$a'_1 = 4$	1	1
$a'_2 = 1$	4	1

Tables $n = 4$.

	a_1	a_2		a_1	a_2	a_3
	6	1		3	3	1
$a'_1 = 6$	1	1	$a'_1 = 3$	2	1	
			$a'_2 = 3$	2	3	
			$a'_3 = 1$	1	1	1

Tables $n = 5$.

	a_1 8	a_2 0	a_3 0	a_4 1	a_1 2	a_2 3	a_3 1	a_4 0		a_1 0	a_2 6	a_3 0	a_4 0
$a'_1 = 8$	1			1	$a'_1 = 2$	1	1			$a'_2 = 6$		5	
$a'_4 = 1$	8			1 ⁶	$a'_2 = 3$	1	3	1					
					$a'_3 = 1$	1	3	3	1*				

Tables $n = 6$.

	a_1 10	a_2 0	a_3 0	a_4 0	a_5 1	a_1 0	a_2 5	a_3 0	a_4 0	a_5 0
$a'_1 = 10$	1				1	$a'_2 = 5$	2			
$a'_5 = 1$	10				1 ⁸	$a'_3 = 2$	4	1		

	a_1	a_2	a_3	a_4	a_5		a_1	a_2	a_3	a_4	a_5
	4	1	3	0	0		3	4	0	1	0
$a'_1 = 4$	1		1			$a'_1 = 3$	2	4		1	
$a'_2 = 1$	4	1	3			$a'_2 = 4$	1	4		1	
						$a'_4 = 1$	2	4		1^2	

Footnote: “Read, ‘Each of the two cubics passes through the point a_4 , the four points a_1 , and, $(1^2, 1)$, twice through one and once through the other of the points a_2 .’”

45. It is to be remarked upon the tables—first, as regards the lines; if we add the numbers in each line, reckoning m^p as mp (that is, each multiple point, according to the number of branches through it), the sums for the successive lines are 2, 5, 8, 11, 14, &c.; that is, each line passes through 2 points, each conic through 5 points, each cubic through 8 points, each quartic through 11 points, &c. But if we add the numbers reckoning m^p as $\frac{1}{2}mp(p+1)$ (that is, each multiple point according to its effect in the determination of the curve), then the sums are 2, 5, 9, 14, 20, &c., that is, all the curves are completely determined, viz., the line by 2 conditions, the conic by 5 conditions, the cubic by 9 conditions, &c. Secondly, as regards the columns, if for any column, reckoning m^p as mp , we multiply each number by the corresponding outside left-hand number, add, and divide the sum by the outside number at the head of the column, the successive results are 2, 5, 8, 11, 14, &c.; this merely expresses the known circumstance that the Jacobian passes $3r-1$ times through each point a_r .

46. The analogous tables showing the passage of the Jacobian through the principal system, in the solutions belonging to certain special forms of n , are

Table $n = p$.

	a_1	a_{p-1}
	$2p-2$	1
$a'_1 = 2p-2$	1	1
$a'_{p-1} = 1$	$2p-2$	1^{p-2}

Tables $n = 2p$.

	a_1	a_2	a_{p-1}	a_p	a_{2p-2}		a_1	a_2	a_{p-1}	a_p	a_{2p-2}
	3	$2p-2$	0	0	1		$2p-2$	1	1	3	0
$a'_1 = 3$	1				1	$a'_2 = 3$		2			
$a'_{2p-2} = 1$	2	$2p-2$			1^{p-2}	$a'_{p-1} =$	$2p-2$	1	1	3	
						$a'_p =$					
						$a'_{2p-1} =$	1	$2p-2$	$p-3$	3^{p-1}	

Tables $n = 2p+1$.

	a_1	a_p	a_{p+1}	a_{2p-1}		a_1	a_p	a_{p+1}	a_{2p-1}
	3	$2p-1$	0	1		$2p-1$	0	3	0
$a'_1 = 3$	1			1	$a'_p = 3$	1			1
$a'_p =$					$a'_p = 2p-1$	1	$2p-1$	1	1
$a'_{p+1} =$					$a'_{p+1} =$				
$a'_{2p-1} =$					$a'_{2p-1} = 1$	$2p-1$	3^{p-1}	1^p	

Tables $n = 3p$.

	a_1	a_2	a_p	a_{2p-1}	a_{3p-2}	a_{3p-1}
	1	4	0	$2p-3$	0	1
$a'_1 = 2p-3$			1			1
$a'_2 = 0$						
$a'_p = 0$						
$a'_p = 4$		3	$2p-3$			1^{p-1}
$a'_{p+1} = 1$	1	4	$2p-3$			1^p
$a'_{2p-1} = 1$	1	4	$(2p-3)^2$			1^{2p-3}
$a'_{2p} =$						

Further tables (book p. 212).

	a_1 $2p-3$	a_2 0	a_p 0	a_{2p-1} 4	a_{3p-1} 1	a_{3p-1} 0
$a'_1 = 1$				1	1	
$a'_2 = 4$			3	1	1	
$a'_p = 2p-3$	1			4	1	1^2
$a'_p = 0$						
$a'_{p+1} = 0$						
$a'_{2p-1} = 0$						
$a'_{3p-2} = 1$	$2p-3$			4^{p-1}	1^p	1^{p-2}
	a_1 4	a_2 1	a_p $2p-2$	a_{p+1} 0	a_{2p} 0	a_{3p-1} 1
$a'_1 = 2p-2$			1			1
$a'_2 = 0$						
$a'_p = 0$						
$a'_{p-1} = 1$			$2p-2$			1^{p-2}
$a'_p = 4$	1	1	$2p-2$			1^{p-1}
$a'_{2p} = 1$	4	1	$(2p-2)^2$			1^{2p-2}
$a'_{3p-1} = 0$						

Final Jacobian table (book p. 213).

	a_1 $2p-2$	a_2 0	a_3 0	a_{p-1} 1	a_{2p} 4	a_{3p-1} 1
$a'_2 = 4$				1	1	1
$a'_3 = 1$					4	1
$a'_{p-1} = 2p-2$	1			1	4	1^p
$a'_{2p} =$						
$a'_{2p-1} =$						
$a'_{3p} =$		$p-2$			4^{p-1}	1^{2p-2}
$a'_{3p-2} = 1$	$2p-2$					

47. The before-mentioned theorem, that $(a_1, a_2, \dots, a_{n-1})$ and $(a'_1, a'_2, \dots, a'_{n-1})$ are the same series of numbers, of course implies $\Sigma a_r = \Sigma a'_r$; this relation Cremona demonstrates independently, by consideration of the pencil of curves

$$(aX + bY + cZ) + \theta(a_1X + b_1Y + c_1Z) = 0,$$

(θ a variable parameter), which corresponds in the first plane to the pencil of lines

$$(ax' + by' + cz') + \theta(a_1x' + b_1y' + c_1z') = 0,$$

which pass through a fixed point $(ax' + by' + cz' = 0, a_1x' + b_1y' + c_1z' = 0)$ in the second figure. In general, in the pencil $U + \theta V = 0$ (U, V given functions of the order n) there are $3(n-1)^2$ values of θ , each giving a nodal curve. But in the present case each of the curves $U = 0, V = 0$ has multiple points at the principal points a_r of the first plane: the question is to obtain the number of values which give a curve having one new double point; and this is found to be

$$= 3(n-1)^2 - \Sigma(r-1)(3r+1)a_r.$$

We have $\Sigma r^2 a_r = n^2 - 1$, $\Sigma r a_r = 3n - 3$; or, substituting, the value of θ is $= \Sigma a_r$. But the curves which have an additional double point are those which correspond to the lines which in the second figure pass through one of the principal points a'_r ; viz., these are the lines drawn from the point $(ax' + by' + cz' = 0, a_1x' + b_1y' + c_1z' = 0)$ to the several principal points a'_r ; and the number of them is $= \Sigma a'_r$. We have thus the required relation $\Sigma a_r = \Sigma a'_r$.

The Quadric Transformation between Two Planes.

48. This is of course given by what precedes. The principal system in each plane is a set of three points; and the Jacobian of the same plane is the set of three lines joining each pair of points; that is, the three lines of either plane are the principal counter-system of the other plane. But to give the analytical investigation directly: taking the coordinates (x, y, z) to refer to the principal system of the first plane (viz., taking the three points to be the vertices of the triangle formed by the lines $x = 0, y = 0, z = 0$), then $X = 0, Y = 0, Z = 0$ being conics through the three points, the functions X, Y, Z will be each of them of the form $fyz + gzx + hxy$; x', y', z' being proportional to three such functions, there will be linear functions of x', y', z' proportional to yz, zx, xy ; or taking these linear functions of the original (x', y', z') for the coordinates (x', y', z') of a point in the second plane, the formulæ of transformation will be

$$x' : y' : z' = yz : zx : xy,$$

and we have then conversely

$$x : y : z = y'z' : z'x' : x'y'.$$

That is, the formulæ for the transformation in question are

$$x' : y' : z' = yz : zx : xy, \quad x : y : z = y'z' : z'x' : x'y'.$$

We at once verify *à posteriori* that the Jacobian in the first plane is $xyz = 0$, and that in the second plane is $x'y'z' = 0$.

The equations may be written

$$x' : y' : z' = \frac{1}{x} : \frac{1}{y} : \frac{1}{z}, \quad x : y : z = \frac{1}{x'} : \frac{1}{y'} : \frac{1}{z'},$$

or, if we please, $xx' = yy' = zz'$; the transformation is thus given as an inverse transformation.

49. With respect to the metrical interpretation and actual construction of the transformation, it is to be observed that if x, y, z be taken to be proportional (not to given multiples of the perpendicular distances, but) to the perpendicular distances of P from the sides of the triangle in the first plane, and similarly x', y', z' to be proportional to the perpendicular distances of P' from the sides of the triangle in the second plane, then in general the equations of transformation must be written, not as above, but in the form

$$\frac{x'}{f} = \frac{y'}{g} = \frac{z'}{h} = \frac{1}{xyz} \text{ a scalar multiple,}$$

involving arbitrary multipliers $f : g : h$. We may imagine in the second plane a point P'' determined by coordinates (x'', y'', z'') , the same coordinates as (x, y, z) , that is, proportional to the perpendicular distances of P'' from the sides of the triangle in the second plane, which point P'' corresponds homographically to P in such wise that

$$\frac{x}{f} : \frac{y}{g} : \frac{z}{h} = x'' : y'' : z''.$$

We have then, in the second plane, the two points P', P'' corresponding to each other in such wise that

$$x'x'' = y'y'' = z'z'';$$

and either of these points being given, the other can at once be constructed; viz., it is obvious that, joining P', P'' with any vertex, say A' , of the triangle $A'B'C'$, the lines $A'P', A'P''$ are equally inclined to the bisectors of the angle A' ; and consequently, P being given, we have the three lines $A'P', B'P'', C'P''$ intersecting in a common point P'' , which is therefore determined by means of any two of these lines. We have thus a geometrical construction of the transformation between P and P'' .

50. The analysis assumes that the principal points A, B, C of the first figure are three distinct points; but they may two of them, or all three, coincide. In the first case, say if B, C coincide, the line BC is still to be regarded as having a definite direction; and taking $x = 0$ for this line, $y = 0$ for the line joining A with (BC) , and $z = 0$ an arbitrary line through A , the functions X, Y, Z will be each of them of the form $\frac{1}{2}fy^2 + \frac{1}{2}gzx + \frac{1}{2}hxy$; and replacing, as before, the original x', y', z' by linear functions of these quantities, these linear functions being taken for the coordinates (x', y', z') , we may write

$$x : y : z = y^2 : xy : xz.$$

Forming the converse system, the equations for the transformation are

$$x' : y' : z' = y^2 : xy : xz, \quad x : y : z = y'^2 : x'y' : x'z';$$

so that the points A', B', C' in the second plane are related as the points in the first plane; viz., B', C' coincide, the line $B'C'$ being definite.

It is easy to verify that the Jacobian in the first plane is $xy^2 = 0$, and the Jacobian in the second plane is $x'y'^2 = 0$.

51. Secondly, if A, B, C all coincide, these being however consecutive points on a curve of finite curvature, or say on a conic; then, taking $x = 0$ for the tangent at (ABC) , $z = 0$ for any other tangent, and $y = 0$ for the chord of contact, the functions X, Y, Z will be of the form $ax^2 + b(y^2 - zx) + 2hxy$; whence we may write

$$x' : y' : z' = x^2 : xy : y^2 - xz.$$

Forming the converse equations, the equations of transformation are

$$x' : y' : z' = x^2 : xy : y^2 - xz, \quad x : y : z = y'^2 - x'y' : x'y' : x'^2.$$

So that the points A', B', C' in the second plane are related as those of the first plane; viz., they are the consecutive points of a curve of continuous curvature.

We may verify that the Jacobian of the first plane is $x^3 = 0$, and the Jacobian of the second plane $x'^3 = 0$.

The Lineo-linear Transformation between Two Planes.

52. We have two equations of the form

$$(a, \dots, x, y, z, x', y', z') = 0, \\ (a_1, \dots, x, y, z, x', y', z') = 0;$$

writing these in the form

$$Px' + Qy' + Rz' = 0, \quad P_1x' + Q_1y' + R_1z' = 0,$$

where (P, Q, R, P_1, Q_1, R_1) are linear functions of (x, y, z) , we have

$$x' : y' : z' \text{ proportional to } \begin{vmatrix} P & Q & R \\ P_1 & Q_1 & R_1 \end{vmatrix},$$

that is to $X : Y : Z$, where $X = 0, Y = 0, Z = 0$ are conics each passing through the same three points in the first plane.

And conversely, writing the equations in the form

$$P'x + Q'y + R'z = 0, \quad P'_1x + Q'_1y + R'_1z = 0,$$

where $(P', Q', R', P'_1, Q'_1, R'_1)$ are linear functions of (x', y', z') , we have

$$x : y : z \text{ proportional to } \begin{vmatrix} P' & Q' & R' \\ P'_1 & Q'_1 & R'_1 \end{vmatrix},$$

that is to $X' : Y' : Z'$, where $X' = 0, Y' = 0, Z' = 0$ are conics each passing through the same three points in the second plane.

53. The lineo-linear transformation is thus the same thing as the quadric transformation. It is, moreover, clear that the equations must, by linear transformations on the two sets of variables respectively, and by linear combination of the two equations, be reducible into forms giving the before-mentioned values of $x : y : z$ and $x' : y' : z'$ respectively. Thus, in the general case, where in each plane the three points are distinct points, the lineo-linear equations will be reducible to

$$xx' - yy' = 0, \quad xx' - zz' = 0;$$

in the case where B, C in the first plane, and B', C' in the second plane respectively coincide, the forms will be

$$xx' - yy' = 0, \quad yz' - y'z = 0;$$

and in the case where A, B, C in the first plane, and A', B', C' in the second plane respectively coincide, the forms will be

$$xy' - yx' = 0, \quad xz' - yy' + zx' = 0.$$

The determination of the actual formulæ for these reductions would, it is probable, give rise to investigations of considerable interest.

The General Rational Transformation between Two Planes (resumed).

54. Consider, as above, the first plane or figure with a principal system $(a_1, a_2, \dots, a_{n-1})$, and the second plane or figure with a principal system $(a'_1, a'_2, \dots, a'_{n-1})$. To a line in the second plane there corresponds in the first plane a curve of the order n passing 1 time through each of the points a_1 , 2 times through each of the points a_2 , 3 times through each of the points a_3 , &c.; or, as we may write this

First figure.	Second figure.
Points $a_1 \ a_2 \ a_3 \ \dots \ a_{n-1}$	Points $a'_1 \ a'_2 \ a'_3 \ \dots \ a'_{n-1}$
1 2 3 ... $n-1$ curve order n	1 curve order 1

viz., the 1's denote the number of times which the curve of the order n passes through the several points a_1 respectively; the 2's the number of times which the curve passes through the several points a_2 respectively; and so on.

55. We may, in the second figure, in the place of a line consider a curve of the order k' . If the equation hereof is $(x', y', z')^{k'} = 0$, then the corresponding curve in the first figure is $(x, y, z)^{k'} = 0$; viz., this is a curve of the order $k = nk'$. If, however, the curve in the second figure passes once or more times through all or any of the points $a'_1, a'_2, \dots, a'_{n-1}$, then there will be a depression in the order of the corresponding curve in the first figure; and, moreover, this curve will pass a certain number of times through all or some of the points $a_1, a_2, \dots, a_{n-1}$. The diagram of the correspondence will be

First figure.	Second figure.
$a_1 \ a_2 \ a_3 \ \dots \ a_{n-1}$	$a'_1 \ a'_2 \ a'_3 \ \dots \ a'_{n-1}$
$a_1 \ a_2 \ a_3 \ \dots \ a_{n-1}$	$a'_1 \ a'_2 \ a'_3 \ \dots \ a'_{n-1}$
$b_1 \ b_2 \ b_3 \ \dots \ b_{n-1}$ curve order k	$b'_1 \ b'_2 \ \dots \ b'_{n-1}$ curve order k'
$c_1 \ c_2 \ \dots$	$c'_1 \ c'_2 \ \dots$
$\vdots \ \dots$	$\vdots \ \dots$

where $a_1, b_1, c_1 \dots$ denote the number of times that the curve of the order k passes through the several points a_1 respectively, (viz., the number of the letters $a_1, b_1, c_1 \dots$ is $= a_1$, any or all of them being zeros,) $a_2, b_2, c_2 \dots$ the number of times that the curve passes through the several points a_2 respectively, (viz., the number of the letters $a_2, b_2, c_2 \dots$ is $= a_2$, any or all of them being zeros,) and so on; and the like for the curve in the second figure.

56. By what precedes, it is easy to see that, if the curve k' passes through a point a'_1 , then the curve k throws off a line, and the depression of order is $= 1$; so, if the curve passes 2 times, 3 times, ... or a'_1 times through the point in question, then the curve throws off the line repeated 2 times, 3 times, ... a'_1 times, or the depression of order is $= 2, 3, \dots$ or a'_1 ; and the like for each of the points a'_1 ; so that, writing for shortness $a'_1 + b'_1 + c'_1 + \dots = \Sigma a'_1$, the depression of order on account of the passages through the several points a'_1 is $= \Sigma a'_1$. Similarly, for each time of passage through a point a'_2 , there is thrown off a conic; or if $a'_2 + b'_2 + \dots = \Sigma a'_2$, then the depression of order is $2\Sigma a'_2$; and so on; and the like for the figure in the other plane; and we thus arrive at the equations

$$\begin{aligned} k &= k'n - \Sigma(a'_1 + 2a'_2 + 3a'_3 \dots + (n-1)a'_{n-1}), \\ k' &= kn - \Sigma(a_1 + 2a_2 + 3a_3 \dots + (n-1)a_{n-1}). \end{aligned}$$

57. The simplest case is when the curve k' does not pass through any of the points $a'_1, a'_2, \dots, a'_{n-1}$. We have then

$$a'_1 = b'_1 = c'_1 = \dots = 0, \quad a'_2 = b'_2 = \dots = 0, \quad \dots \quad a'_{n-1} = b'_{n-1} = \dots = 0;$$

consequently $k = k'n$. And, moreover, it is easy to see that

$$a_1 = b_1 = \dots = k', \quad a_2 = b_2 = \dots = 2k', \quad \dots \quad a_{n-1} = b_{n-1} = \dots = (n-1)k';$$

so that the correspondence is:

First Figure.	Second Figure.
$k' \ 2k' \ 3k' \ \dots \ (n-1)k'$	0
$k' \ 2k' \ 3k' \ \dots \ (n-1)k'$ curve order $k = nk'$	k'
$\vdots$	curve order k'

We have

$$\Sigma a_1 = k'a_1, \quad \Sigma a_2 = 2k'a_2, \quad \frac{1}{2}a_1(a_1 - 1) = k'(k' - 1)a_1, \text{ \&c.},$$

and the formulæ for k, k' become

$$k = k'n, \quad k' = kn - k'(a_1 + 4a_2 + \dots + (n-1)^2 a_{n-1});$$

viz., the second equation is here $k' = kn - k'(n^2 - 1)$; that is, $k'n^2 = kn$, agreeing, as it should do, with the first equation.

58. Moreover, the deficiency-relation is

$$\frac{1}{2}(k-1)(k-2) - \Sigma \frac{1}{2}[k'(k'-1) + 2k'(2k'-1) \dots + (n-1)k'((n-1)k'-1)] = \frac{1}{2}(k'-1)(k'-2);$$

or, what is the same thing, this is

$$(nk' - 1)(nk' - 2) - (k' - 1)(k' - 2) = k'^2\{a_1 + 4a_2 + \dots + (n-1)^2 a_{n-1}\} - k'\{a_1 + 2a_2 + \dots + (n-1)a_{n-1}\}.$$

The right-hand side is

$$k'^2(n^2 - 1) - k'(3n - 3) = (n-1)k'\{(n+1)k' - 3\},$$

and we have thus the identical equation

$$(nk' - 1)(nk' - 2) - (k' - 1)(k' - 2) = (n-1)k'\{(n+1)k' - 3\}.$$

59. It should be possible, when the nature of the correspondence between the two planes is completely given, to express each of the numbers $a_1, b_1, c_1, \dots, a_{n-1}, b_{n-1}, \dots$ in terms of $k', a'_1, b'_1, c'_1, \dots, a'_{n-1}, b'_{n-1}, \dots$; and reciprocally each of the numbers $a'_1, b'_1, c'_1, \dots, a'_{n-1}, b'_{n-1}, \dots$ in terms of $k, a_1, b_1, c_1, \dots, a_{n-1}, b_{n-1}, \dots$, thus completing a system of relations between the two sets

$$(k, a_1, b_1, c_1, \dots, a_{n-1}, b_{n-1}, \dots), \quad (k', a'_1, b'_1, c'_1, \dots, a'_{n-1}, b'_{n-1}, \dots);$$

but even if the theory was known, there would be considerable difficulty in forming a proper algorithm for the expression of these relations.

60. The two curves must have each of them the same deficiency. It is to be noticed, that if the curve in the first plane passes any number of times through a point P , which is not one of the points $a_1, a_2, \dots, a_{n-1}$, then the corresponding curve in the second plane will pass the same number of times through the corresponding point P' , which point will not be one of the points $a'_1, a'_2, \dots, a'_{n-1}$. The points P, P' will therefore contribute equal values to the deficiencies of the two curves respectively; so that, in equating the two deficiencies, we may disregard P, P' , and attend only to the points $a_1, a_2, \dots, a_{n-1}$ of the first plane, and $a'_1, a'_2, \dots, a'_{n-1}$ of the second plane. The required relation thus is

$$\begin{aligned} & \frac{1}{2}(k-1)(k-2) - \frac{1}{2}\{a_1(a_1-1) + a_2(a_2-1) + \dots + a_{n-1}(a_{n-1}-1)\} \\ &= \frac{1}{2}(k'-1)(k'-2) - \frac{1}{2}\{a'_1(a'_1-1) + a'_2(a'_2-1) + \dots + a'_{n-1}(a'_{n-1}-1)\}. \end{aligned}$$

61. In the case of the quadric transformation $n = 2$, we have in the first plane the three points a_1 , say these are A, B, C ; and in the second plane the three points a'_1 , say these are A', B', C' . And if in the first plane the curve of the order k passes a, b, c times through the three points respectively, and in the second plane the corresponding curve of the order k' passes a', b', c' times through the three points respectively, then it is easy to obtain

$$\begin{aligned} k' &= 2k - a - b - c, & k &= 2k' - a' - b' - c', \\ a' &= k - b - c, & a &= k' - b' - c', \\ b' &= k - c - a, & b &= k' - c' - a', \\ c' &= k - a - b, & c &= k' - a' - b'. \end{aligned}$$

The Quadric Transformation any number of times repeated.

62. We may successively repeat the quadric transformation according to the type

First Fig.	Second Fig.	Third Fig.	Fourth Fig.
A, B, C	A', B', C'	A'', B'', C''	A''', B''', C'''
	D', E', F'	D'', E'', F''	D''', E''', F'''
		G'', H'', I''	G''', H''', I'''

viz., in the transformation between the first and second figures, the principal systems are ABC and $A'B'C'$ respectively; in that between the second and third figures, they are $D'E'F'$ and $D''E''F''$ respectively; in that between the third and fourth figures, they are $G''H''I''$ and $G'''H'''I'''$; and so on. And it is then easy to see that between the first and any subsequent figure we have a rational transformation of the order 2 for the second figure, 4 for the third figure, 8 for the fourth figure, and so on.

63. But to further explain the relation, we may complete the diagram, by taking, in the transformation between the second and third figures, A'', B'', C'' to correspond to A', B', C' ; similarly, in that between the third and fourth, A''', B''', C''' to correspond to A'', B'', C'' ; and D''', E''', F''' to correspond to D'', E'', F'' . And so in the transformation between the second and third figure, we may make G', H', I' to

correspond to G'', H'', I'' , and between the first and second figures make D, E, F to correspond to D', E', F' , and G, H, I to G', H', I' , the diagram being thus:

First Fig.	Second Fig.	Third Fig.	Fourth Fig.
A, B, C	A', B', C'	A'', B'', C''	A''', B''', C'''
D, E, F	D', E', F'	D'', E'', F''	D''', E''', F'''
G, H, I	G', H', I'	G'', H'', I''	G''', H''', I'''

Observe that in the principal systems (for instance, A, B, C and A', B', C') the points A, B, C correspond, not to the points A', B', C' , but to the lines $B'C', C'A', A'B'$ respectively; and so in the other case.

64. Consider now a line in the first figure: there corresponds hereto in the second figure a conic through the points A', B', C' ; and to this conic there corresponds in the third figure a quartic curve passing through each of the points A'', B'', C'' once, and through each of the points D'', E'', F'' twice. And conversely, to a line in the third figure corresponds in the second figure a conic through the points D', E', F' ; and hereto in the first figure a quartic through the points D, E, F , once and through the points A, B, C twice; that is, we have between the first and third figures a quartic transformation wherein $a_1 = a_2 = 3$ and $a'_1 = a'_2 = 3$, or say a quartic transformation $3_1 3_2$ and $3_1 3_2$. In like manner, passing to the fourth figure, to a line in the first figure corresponds in the fourth figure an octic curve passing through A''', B''', C''' once, through D''', E''', F''' twice, and through G''', H''', I''' four times; and conversely, to a line in the fourth figure there corresponds in the first figure an octic curve passing through the points G, H, I once, the points D, E, F twice, and the points A, B, C four times; that is, between the first and fourth figures we have an octic transformation, wherein $a_1 = a_2 = a_4 = 3$, $a'_1 = a'_2 = a'_4 = 3$, or say a transformation, order 8, of the form $3_1 3_2 3_4$ and $3_1 3_2 3_4$. And so between the first and fifth figures there is a transformation, order 16, of the form $3_1 3_2 3_4 3_8$ and $3_1 3_2 3_4 3_8$.

65. It is, moreover, easy to find the Jacobians or counter-systems in the several transformations respectively. Thus, in the transformation between the first and second figures, in the second figure the Jacobian consists of 3 lines such as $B'C'$ (viz., these are, of course, the lines $B'C', C'A', A'B'$). Hence, in the transformation between the first and third figures, the Jacobian in the third figure consists of

3 conics $B''C'' (D''E''F'')$,
3 lines $D''E''$;

viz., one of the conics is that through the five points B'', C'', D'', E'', F'' , one of the lines that through the two points D'', E'' . And so in the fourth figure, the Jacobian consists of

3 quartics $B'''C''' (D'''E'''F''') (G'''H'''I''')$,
3 conics $D'''E''' (G'''H'''I''')$,
3 lines $G'''H'''$;

viz., one of the quartics passes through B''', C''' ; through D''', E''', F''' each once; and through G''', H''', I''' each twice. And so in the fifth figure the Jacobian consists of

3 octics $B''''C'''' (D''''E''''F'') (G''''H''''I'') (J''''K''''L'')$,
3 quartics $D''''E'''' (G''''H''''I'') (J''''K''''L'')$,
3 conics $G''''H'''' (J''''K''''L'')$,
3 lines $J''''K''''$,

and so on.

66. The conditions are in each case sufficient for the determination of the curve. This depends on the numerical relation

$$4 + 3\{1 \cdot 2 + 2 \cdot 3 + 4 \cdot 5 + 8 \cdot 9 + \dots + 2^q(2^q - 1)\} = \frac{1}{2} \cdot 2^{q+1}(2^{q+1} + 3).$$

The term in $\{ \}$ is

$$1 + 4 + 16 + \dots + 2^{2q} + 1 + 2 + 4 + \dots + 2^q,$$

that is

$$\frac{2^{2q+2} - 1}{2^2 - 1} + \frac{2^{q+1} - 1}{2 - 1},$$

which is

$$= \frac{1}{3}[2^{2q+2} + 3 \cdot 2^{q+1} - 4];$$

and the relation is thus identically true.

67. Conversely, in the transformation between the first figure and the several other figures respectively, the Jacobian of the first figure is

3 lines AB and so on; for order 2, between first and second figures;
 3 conics $DE(ABC)$,
 3 lines AB for order 4, between first and third figures;
 3 quartics $GH(DEF)(ABC)$,
 3 conics $DE(ABC)$,
 3 lines AB for order 8, between first and fourth figures;
 3 octics $JK(GHI)(DEF)(ABC)$,
 3 quartics $GH(DEF)(ABC)$,
 3 conics $DE(ABC)$,
 3 lines AB for order 16, between first and fifth figures;

and so on.

Special Cases—Reduction of the General Rational Transformation to a Series of Quadric Transformations.

68. It was remarked by Mr Clifford that any Cremona-transformation whatever may be obtained by this method of repeated quadric transformations, if only the principal systems, instead of being completely arbitrary, are properly related to each other. To take the simplest instance; suppose that we have

First figure.	Second figure.	Third figure.
A, B, C	A', B', C'	B'', C''
		D'', E'', F''
E, F	$D' = A', E', F'$	

viz., in the transformation between the first and second figures, we have the principal systems ABC and $A'B'C'$ (arbitrary as before); but in the transformation between the second and third figures, the principal systems are $D'E'F'$ and $D''E''F''$, where D' , instead of being arbitrary, coincides with A' . And we then have B'', C'' in the third figure corresponding to B', C' in the second figure, and E, F in the first figure corresponding to E', F' in the second figure. This being so, to a line in the first figure corresponds in the second figure a conic through A', B', C' . But $A' = D'$; viz., this conic passes through a point D' of the principal system of the second figure, in regard to the transformation between the second and third figures. That is, (k, a, b, c referring to the second figure, and k', a', b', c' to the third figure, $k = 2, a = 1, b = 0, c = 0$, and therefore $k' = 3, a' = 2, b' = 1, c' = 1$.) corresponding to the conic we have in the third figure a curve, order 3 (cubic curve), passing twice through D'' , but once through E'' and F'' respectively; this cubic curve passes also through the points B'', C'' which correspond to B', C' respectively; that is,

cubic passes through	E'', F'', B'', C''	each 1 time
„ „	D''	2 times

or, corresponding to a line in the first figure, we have in the third figure a curve, order 3, passing through four fixed points each 1 time, and through one fixed point 2 times. That is, we have $n = 3, a'_1 = 4, a'_2 = 1$. And in the same manner, to a line in the third figure there corresponds in the first figure a cubic through four fixed points (viz., B, C, E, F) each 1 time, and through one fixed point, A , 2 times; so that also $a_1 = 4, a_2 = 1$. The transformation is thus of the order 3, and the form $4_1 1_2$ and $4_1 1_2$ (this is in fact the only cubic transformation; see the Tables, *ante*, No. 41).

69. Mr Clifford has also devised a very convenient algorithm for this decomposition of a transformation of any order into quadric transformations. The quadric transformation is denoted by [3], the cubic transformation by [41], the quartic transformations by [601], [330], the quintic ones by [8001], [3310], [0600], and so on; see the Tables just referred to. (This is substantially the same as a notation employed above, the zeros enabling the omission of the suffixes; viz., [8001] = $8_1 1_4$; and so in other cases.)

70. The foregoing result is represented thus $[4, 1] = [3\$0, 0, 1]$, which I proceed to explain. Consider in the first figure a line; the symbol [3] denotes that in the second figure we have a conic with three points (a'_1). We are about to apply to this a quadric transformation (0, 0, 0); would denote that the three points of the principal system in the second figure were all of them arbitrary; (0, 0, 1) that one of these points was a point a'_1 ; (0, 1, 1) that two of them were points a'_1 ; (1, 1, 1) that all three of them were points a'_1 ; (0, 0, 2) would denote that one of the points was a point a'_2 ; only in the present case we can have no such symbol, by reason that there are no points a'_2 . Hence [3\$001] denotes that the conic has applied to it a quadric transformation such that, in the transformation thereof, one point of the principal system coincides with one of the points (a'_1) on the conic. To [3], qua quadric transformation, belongs the number 2; and from 2, (001) we derive 3, (112), (in general $k, (a, b, c)$ gives $k', (a', b', c')$, where $k' = 2k - a - b - c$, $a' = k - b - c$, $b' = k - c - a$, $c' = k - a - b$). $k = 2$ corresponds to a symbol [3] of one number, $k' = 3$ to a symbol of two numbers; viz., we change [3] into [30]; we then, in the symbols (112) and (001), consider the frequencies of the several numbers 1, 2, ... taking those in the first symbol as positive, and those in the second symbol as negative; or, what is the same thing, representing the frequency as an index, we have $1^2 2^1 1^{-1}$; or, combining, $1^{2-1} 2^1$; these indices are then added on to the numbers of [30]; viz., the index of 1 to the first number, the index of 2 to the second number (and, in the case of more numbers, so on); [30] is thus converted into [41], and we have the required equation

$$[41] = [3\$001],$$

where the rationale of this algorithmic process appears by the explanation, *ante*, No. 68.

71. As another example take [8001] = [601\$003]. To [601], *quâ* quartic transformation, belongs the number 4; and from 4, (003) we form 5, (114); where the 5 indicates that [601] is to be changed into [6010]; then (114), (003), writing them in the form $1^2 2^0 3^{-1} 4^1$, show that to the numbers of [6010] we are to add 2, 0, -1, 1; thus changing the symbol into [8001], so that we have the required relation.

72. Mr Clifford calculated in this way the following table, showing how any transformation of an order not exceeding 8 can be expressed by means of a series of quadric transformations; the symbols Cr. 3, Cr. 4.1, 4.2, &c., refer to the order and number of Cremona's tables, *ante*, No. 41.

(Cremona / Clifford reduction table, books pp. 223–224.)

$$\begin{aligned}
 \text{Cr. 3. } [41] &= [3001], \\
 \text{Cr. 4.1 } [601] &= [41002] = [3001002], \\
 4.2 [330] &= [3000] = [41011] = [3001011], \\
 \text{Cr. 5.1 } [8001] &= [601003] = [3001002003], \\
 5.2 &= [3310] \\
 5.3 [0600] &= [41001] = [3001001], \\
 &= [330111] \\
 \text{Cr. 6.1 } [10, 0001] &= [3303111] = [3001011111], \\
 6.2 [14200] &= [8001004] \\
 6.3 [41300] &= [330011] = [3001002003]004, \\
 &= [601011] \\
 6.4 [34010] &= [3310022] = [3001011] = [3001011011], \\
 &= [3303002] \\
 &= [3310013] = [3001002011] \\
 &= [3001001022] \\
 &= [3000002] \\
 &= [3001001013] \\
 \text{Cr. 7.1 } [12, 00001] &= [10, 0001005] = [3001002003004005], \\
 7.2 [330001] &= [3000001] = [232100], \\
 7.3 [034000] &= [3310111] = [3001001111], \\
 7.4 [503100] &= [601001] = [3001002001], \\
 7.5 [350010] &= [3310003] = [3001001003], \\
 \text{Cr. 8.1 } [14, 000001] &= [3001002003004005006], \\
 8.2 [3230100] &= [3310002] = [3001001002], \\
 8.3 [1322000] &= [3310011] = [3001001011], \\
 8.4 [0070000] &= [0340222] = [3001001111222], \\
 8.5 [3600010] &= [34010004] = [3303002004] = [3000002004], \\
 8.6 [6013000] &= [601000] = [3001002000], \\
 8.7 [0520100] &= [0600002] = [3000111222], \\
 8.8 [2051000] &= [4130112] = [3001000112], \\
 8.9 [3303000] &= [3000000000].
 \end{aligned}$$

73. The reduction as above of a transformation to a series of quadric transformations, enables the determination of the reciprocal transformation; or, what is the same thing, the determination of the Jacobian of the first figure; see the example, *ante*, No. 67, where it appears that the reciprocal transformation of [41] is [41]. But I do not see any easy algorithmic process for the determination of the reciprocal transformation, or still less any general form in which the result can be expressed; and I do not at present pursue the inquiry.

The Rational Transformation between Two Spaces.

74. The general principles have been already explained: the two systems

$$x : y : z : w = X : Y : Z : W \quad \text{and} \quad x : y : z : w = X' : Y' : Z' : W'$$

must be derivable the one from the other; and starting with the first system, this will be the case if only the surfaces $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ have a common intersection equivalent to $n^3 - 1$ points of intersection, but not equivalent to a complete common intersection of n^3 points. The last-mentioned circumstance would arise, if the condition of the common intersection should impose upon the surface more than $\frac{1}{6}(n+1)(n+2)(n+3) - 4$ conditions; viz., the surfaces would then be connected by an identical equation or syzygy $\alpha X + \beta Y + \gamma Z + \delta W = 0$. The common intersection is a figure composed of points and curves: say it is the principal system in the first space; the problem is, to determine a principal system equivalent to $n^3 - 1$ points of intersection but such that the number of conditions to be satisfied by a surface passing through it is not more than

$$\frac{1}{6}(n+1)(n+2)(n+3) - 4.$$

75. The following locutions are convenient. We may say that the number of conditions imposed upon a surface of the order n which passes through the common intersection is the *Postulation* of this intersection; and that the number of points represented by the common intersection (in regard to the points of intersection of any three surfaces each of the order n which pass through it) is the *Equivalence* of this intersection. The conditions above referred to are thus

$$\text{Equivalence} = n^3 - 1,$$

$$\text{Postulation} \not\asymp \frac{1}{6}(n+1)(n+2)(n+3) - 4.$$

76. It would appear by the analogy of the rational transformation between two planes, that the only cases to be considered are those for which

$$\text{Postulation} = \frac{1}{6}(n+1)(n+2)(n+3) - 4;$$

but I cannot say whether this is so.

77. In the transformation between two planes, the two conditions lead, as was seen, to the result that the curve $aX + bY + cZ = 0$ is unicursal. I do not see that in the present case of two spaces, the two conditions lead to the corresponding result that the surface $aX + bY + cZ + dW = 0$ is unicursal; that this is so, appears, however, at once from the general notion of the rational transformation. In fact, the equation in question $aX + bY + cZ + dW = 0$ is satisfied by $x : y : z : w = X' : Y' : Z' : W'$ and $ax' + by' + cz' + dw' = 0$; the last equation determines the ratios $x' : y' : z' : w'$ in terms of two arbitrary parameters (say these are $x' : y'$ and $x' : z'$), and we have then $x : y : z : w$ proportional to rational functions of these two parameters; that is, the surface $aX + bY + cZ + dW = 0$ is unicursal. And similarly the surface $aX' + bY' + cZ' + dW' = 0$ is unicursal.

78. In the most general point of view, the principal system will contain a given number of points which are simple points, a given number which are quadric points, a given number which are cubiconical points, &c. &c., on the surfaces; and similarly a given number of curves which are simple curves, a given number which are double curves, &c. &c., on the surfaces. But, to simplify, I will consider that it includes only points which are simple points, and a curve which is a simple curve on the surfaces: this curve may, however, break up into separate curves, and we thus, in fact, admit the case where there are any number of separate curves—each of them a simple curve on the surfaces. It is right to remark that we cannot assert *a priori*—and it is not in fact the case—that the principal system in the second space will be subject to the like restrictions: starting with such a principal system in the first space, we may be led in the second space to a principal system including a curve which is a double curve on the surfaces: an instance of this will in fact occur.

79. It is shown (Salmon's *Solid Geometry*, 2nd ed., p. 283; [Ed. 4, p. 321]), that in the intersection of three surfaces of the orders μ, ν, ρ respectively, a curve of intersection of the order m and class r counts as $m(\mu + \nu + \rho - 2) - r$ points of intersection. For a curve without actual double points or stationary points, we have $r = m(m - 1) - 2h$, where h is the number of apparent double points; or, substituting, we have the curve counting for $m(\mu + \nu + \rho - 2) - m(m - 1) + 2h$ points of intersection; this is in fact a more general form of the formula, inasmuch as it extends to the case of a curve with actual double points and stationary points. Or, what is the same thing, the three surfaces intersecting in the curve of the order m with h apparent double points, will besides intersect in $\mu\nu\rho - m(\mu + \nu + \rho - 2) + m(m - 1) - 2h$ points; viz., the curve may, besides the apparent double points, have actual double points and stationary points; but these do not affect the formula.

80. Some caution is necessary in the application of the theorem. For instance, to consider cases that will present themselves in the sequel: let the surfaces be cubics ($\mu = \nu = \rho = 3$); the number of remaining intersections is given as $27 - 7m + m(m - 1) - 2h$. Suppose that the curve consists of four non-intersecting lines, $m = 4$, $h = 6$, the number is given as -1 . But observe in this case there are two lines each meeting the four given lines; that is, any cubic surface passing through the four given lines meets these two lines each of them in four points, that is, the cubic passes also through each of the two lines; the complete *curve*-intersection of the surfaces is made up of the six lines $m = 6$, $h = 7$ (since each of the two lines, as intersecting the four lines, gives actual double points, but the two lines together give one apparent double point), and the expression for the number of the remaining points of intersection becomes $27 - 42 + 30 - 14 = 1$, which is correct.

81. Similarly, if the given curve of intersection be a conic and two non-intersecting lines, there is here in the plane of the conic a line meeting each of the two given lines, and therefore meeting the cubic surface, in four points, that is, lying wholly in the cubic surface: the complete *curve*-intersection consists of the conic, the two given lines, and the last-mentioned line, $m = 5$, $h = 5$, and the number of points of intersection $= 27 - 35 + 20 - 10 = 2$, which is correct. Again, if the given curve of intersection be two conics, here the line of intersection of the planes of the conics lies in the cubic surface; or, for the complete *curve*-intersection we have $m = 5$, $h = 4$; and the number of points is $27 - 35 + 20 - 8 = 4$. If in this last case each or either of the conics become a pair of intersecting lines, or if in the preceding case the conic becomes a pair of intersecting lines, the results remain unaltered.

82. If a surface of the order μ pass through a curve of the order m and class r without stationary points or actual double points, this imposes on the surface a number of conditions $= m(\mu + 1) - \frac{1}{2}r$. In the case in question, the value of r is $= m(m - 1) - 2h$; or, substituting, the number of conditions is $= (\mu + 1)m - \frac{1}{2}m(m - 1) + h$; and the formula in this form holds good even in the case where the curve has stationary points and actual double points. Thus $\mu = 3$, the number of conditions is $4m - \frac{1}{2}m(m - 1) + h$. If the curve be a line, $m = 1$, $h = 0$, number of conditions is $= 4$; if the curve be a pair of non-intersecting lines, $m = 2$, $h = 1$, number of conditions is $= 8$. And so in general, if the curve consist of k non-intersecting lines ($k = 4$ at most), then $m = k$, $h = \frac{1}{2}k(k - 1)$, and the number of conditions is $= 4k$. If the curve be a conic, or a pair of intersecting lines, $m = 2$, $h = 1$, and the number of conditions is $= 7$. If the curve consist of k lines, such that there are θ pairs of intersecting lines, then $m = k$, $h = \frac{1}{2}k(k - 1) - \theta$, and the number of conditions is $= 4k - \theta$. It is obvious that, the number of conditions for a line being $= 4$, that for the k lines with θ intersecting pairs must have the foregoing value $-\theta$. In fact, when the lines do not intersect, we take on each line 4 points, and the cubic surface passing through any such 4 points will contain the line; but for two lines which intersect, taking this point, and on each of the intersecting lines 3 other points, the cubic surface through the 7 points will pass through the two lines; and so in other cases.

83. The formula must, in some instances, be applied with caution. Thus, given five non-intersecting lines $k = 5$, $\theta = 0$, and the number of conditions is $= 20$; and a cubic surface cannot be, in general, made to pass through the lines. But if the five lines are met by any other line, then a cubic surface, if it pass through the five lines, will pass through this sixth line; for the six lines $k = 6$, $\theta = 5$, and the number of conditions is $24 - 5 = 19$; so that there is a determinate cubic surface through the six lines, and therefore through the five lines related in the manner just referred to.

84. Recurring to the problem of transformation, it appears by what precedes, that if the principal system in the first plane consists of a_1 points, and of a curve of the order m_1 with h_1 apparent double points (the a_1 points being simple points, and the curve a simple curve on the surfaces), then the conditions for a transformation are

$$(3n - 2)m_1 - m_1(m_1 - 1) + 2h_1 + a_1 = n^3 - 1,$$

$$(n + 1)m_1 - \frac{1}{2}m_1(m_1 - 1) + h_1 + a_1 = \frac{1}{6}(n + 1)(n + 2)(n + 3) - 4,$$

where, in the second line, instead of $\neq$ I have written $=$. I remark, in passing, that I have ascertained that an actual triple point counts as an apparent double point; or, what is the same thing, that if the curve has

t_1 actual triple points, then we may, instead of h_1 , write $h_1 + t_1$. The equations give

$$m_1(4n - 5 - m_1) = \frac{1}{3}(n - 1)(5n^2 - n - 12) - 2h_1,$$

$$(n - 4)m_1 - a_1 = \frac{1}{3}(n - 1)(2n^2 - 4n - 15),$$

to which may be joined

$$(3n + 8)m_1 - 2m_1(m_1 - 1) + 4h_1 + 5a_1 = (n - 1)(6n + 17).$$

The first two equations for the successive values of n give

$$\begin{array}{lll} n = 2, & m_1(3 - m_1) = 2 - 2h_1, & 2m_1 + a_1 = 5 \\ n = 3, & m_1(7 - m_1) = 20 - 2h_1, & m_1 + a_1 = 6 \\ n = 4, & m_1(11 - m_1) = 64 - 2h_1, & a_1 = -1 \\ n = 5, & m_1(15 - m_1) = 144 - 2h_1, & -m_1 + a_1 = -20 \\ n = 6, & m_1(19 - m_1) = 270 - 2h_1, & -2m_1 + a_1 = -55 \\ \&c. & \&c. & \&c. \end{array}$$

85. It is remarkable that for $n = 4$ there is no solution of the geometrical problem; in fact, $a_1 = -1$, a negative value of a_1 , shows that this is so. For the higher values of n , there seem to be solutions with large values of m_1, h_1, a_1 ; for example, $n = 5$, we have $m_1 = 20 + a_1$, is $= 20$ at least. Writing $m_1 = 20$, we have $-100 = 144 - 2h_1$, or $2h_1 = 244$. The highest value of $2h_1$ is $= (m_1 - 1)(m_1 - 2)$, which for $m_1 = 20$ is $= 342$; or the foregoing value $2h_1 = 244$ is admissible. Thus $m_1 = 20, h_1 = 122, a_1$ gives a solution; and, moreover, any larger value of m_1 , say $m_1 = 20 + \alpha$, gives an admissible solution, $m_1 = 20 + \alpha, h_1 = 122 + \frac{1}{2}\alpha(\alpha + 25), a_1 = \alpha$. And so for $n = 6$, &c.; but I have not further examined any of these cases, and do not understand them.

There remain the cases $n = 2, n = 3$. For $n = 2$, since $2m_1 + a_1 = 5$, we have $m_1 = 0, 1$, or 2 ; $m_1 = 0$ gives $h_1 = 0$, which is not admissible. The remaining solutions are $m_1 = 1, h_1 = 0, a_1 = 3$; and $m_1 = 2, h_1 = 0, a_1 = 1$.

For $n = 3$, since $m_1 + a_1 = 6$, we have $m_1 = 0, 1, 2, 3, 4, 5$, or 6 . $m_1 = 0$ gives $h_1 = 10$; $m_1 = 1$ gives $h_1 = 7$; $m_1 = 2$ gives $h_1 = 5$; $m_1 = 3$ gives $h_1 = 4$: these values are not geometrically admissible. The remaining cases are $m_1 = 4, h_1 = 4, a_1 = 2$; $m_1 = 5, h_1 = 5, a_1 = 1$; $m_1 = 6, h_1 = 7, a_1 = 0$.

86. The reciprocal transformation is in every case of the order $n' = n^2 - m_1$. Hence considering the quadric transformations:

First, the case $n = 2, m_1 = 1, h_1 = 0, a_1 = 3$; the reciprocal transformation is of the order $n' = 3$. Suppose for a moment that the principal system in the second space is of the same nature as that above considered in the first space, consisting of a'_1 points, and a curve of the order m'_1 with h'_1 apparent double points (the a'_1 points each a simple point, and the curve a simple curve on the surfaces $X' = 0$, &c.). Passing back to the original transformation, we should have $2 = \frac{9 - m'_1}{1}$, that is, $m'_1 = 7$. But it has just been seen that, for $n = 3$, the only values of m_1 are $4, 5, 6$; hence for $n' = 3$ we cannot have $m'_1 = 7$. The explanation is, that the principal system in the second space is not of the form in question; it, in fact, consists (as will appear) of three lines each a simple line, and of another line which is a double line on the surfaces $X' = 0$, &c. In the intersection of any two of these surfaces, the three lines count each once, the double line four times, and the order of the curve of intersection is thus $3 + 4 = 7$, as it should be. The principal system may be characterized $a'_1 = 0, m'_1 = 3, \nu = 3, h'_1 = 0$, where here ν is the number of simple lines and a separate count is made of the double line.

(End of chunk: PDF pages 226–250, book pages 204–228. Paper 447, Cremona-tables of article 41 through the opening of Article 86. The text resumes in the next chunk at the detailed enumeration of the principal systems in three-space.)